

Update Potensiostat

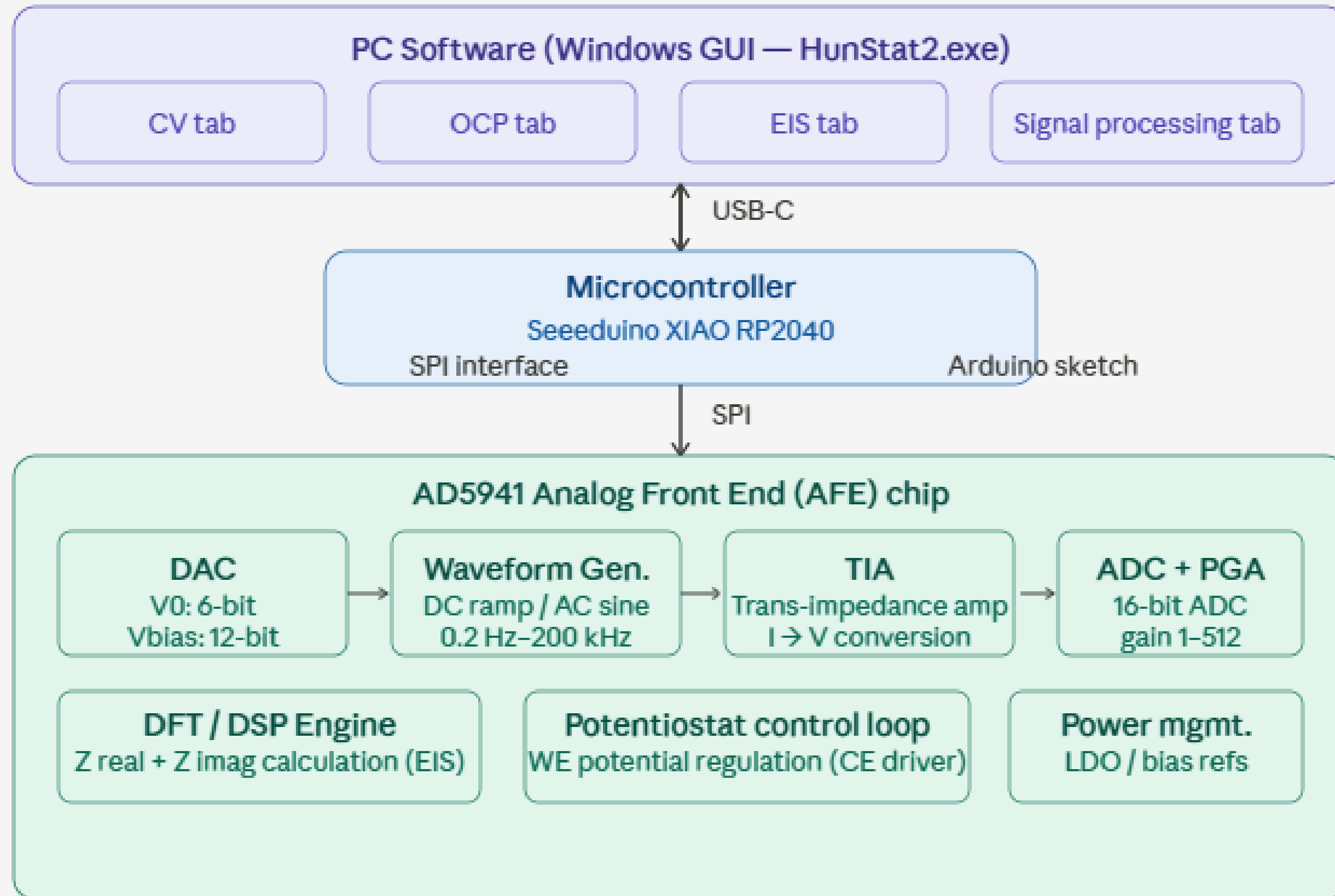
Dipresentasikan oleh

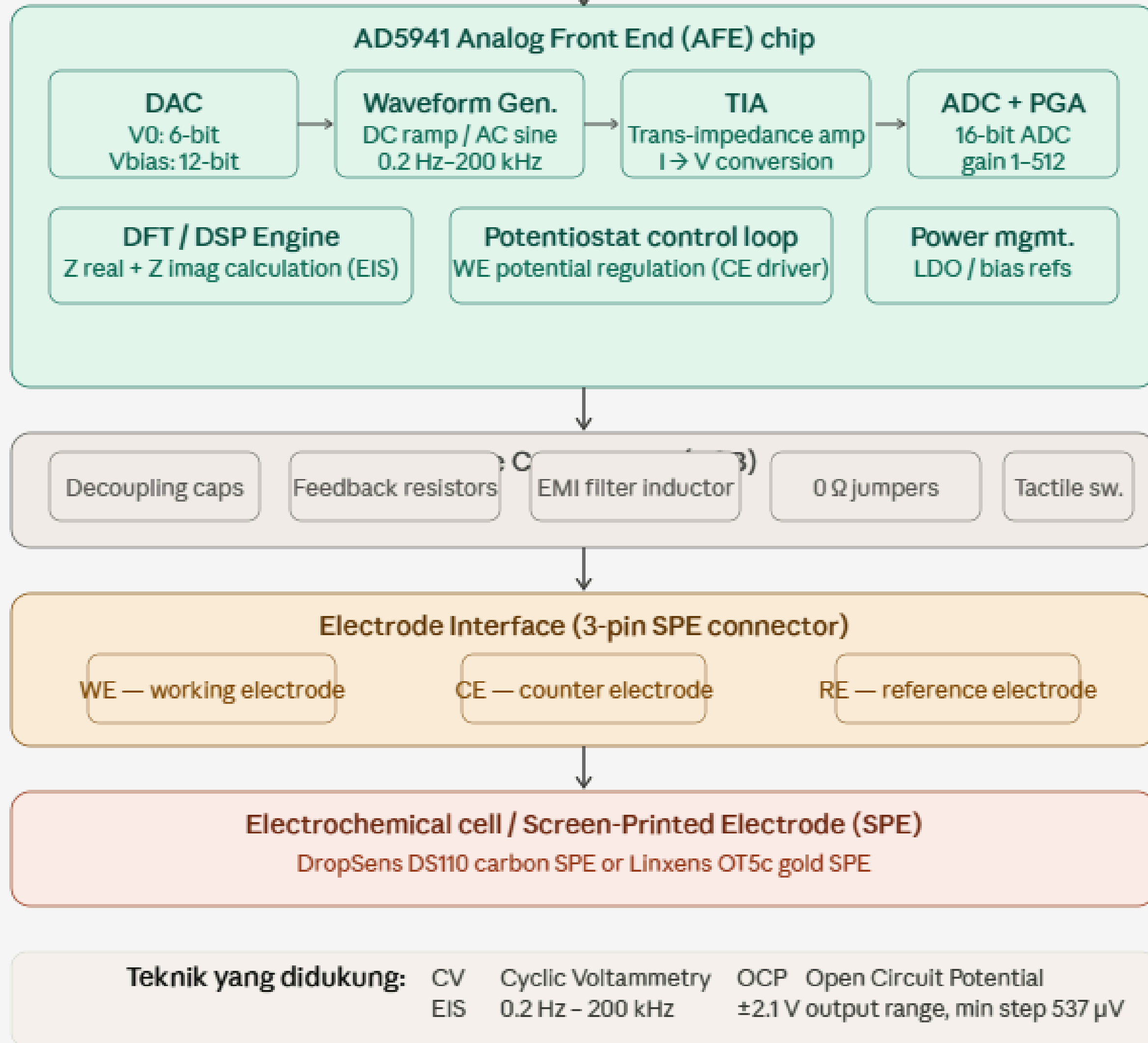
30 MEI 26

Daftar Isi:

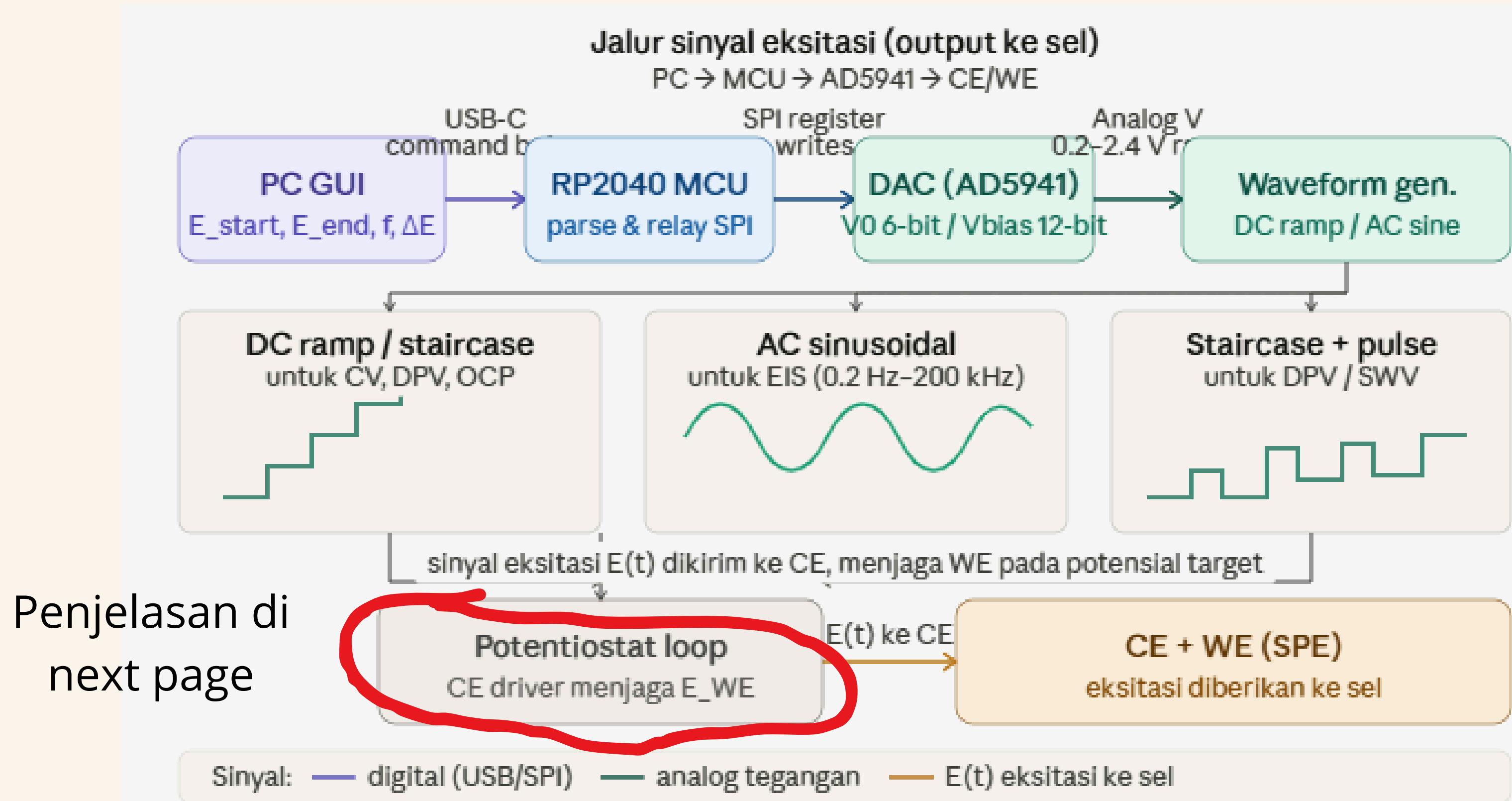
- 1. Dokumen (diagram blok, RCAL)**
- 2. Foldering Software**
- 3. Laporan jasa Potensiostat**

1. Dokumen (Diagram Blok)

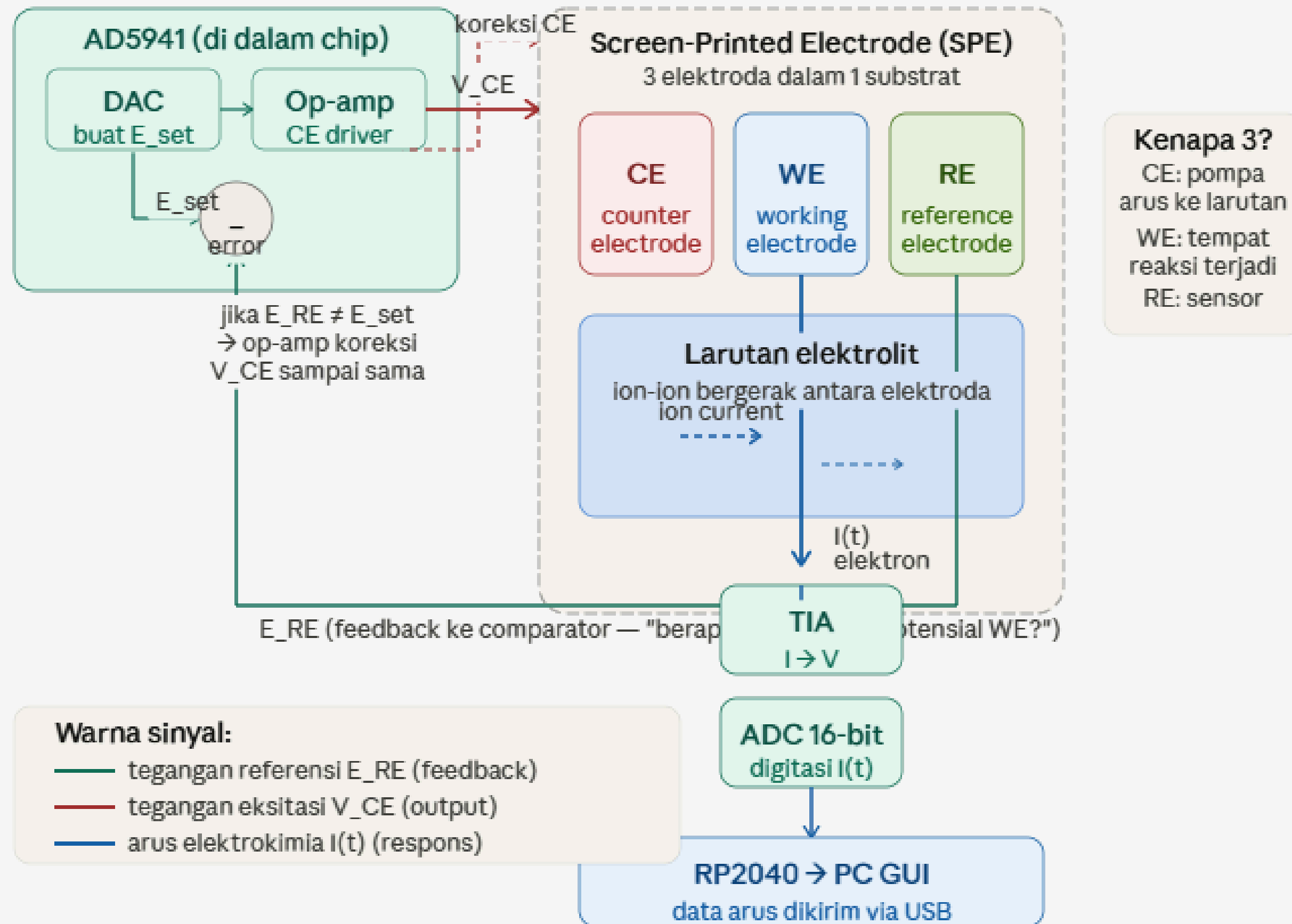


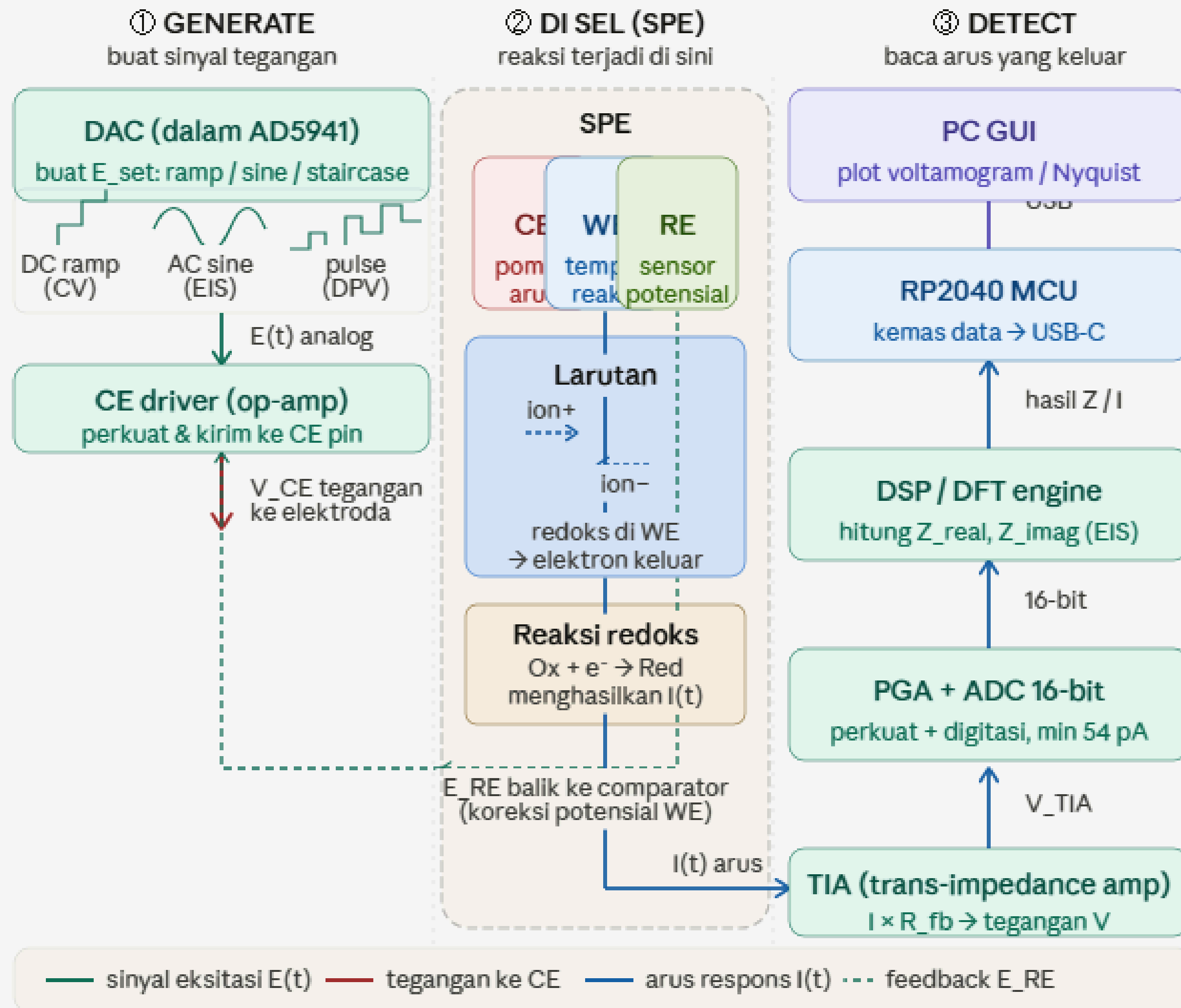


1. Dokumen (Diagram Blok)

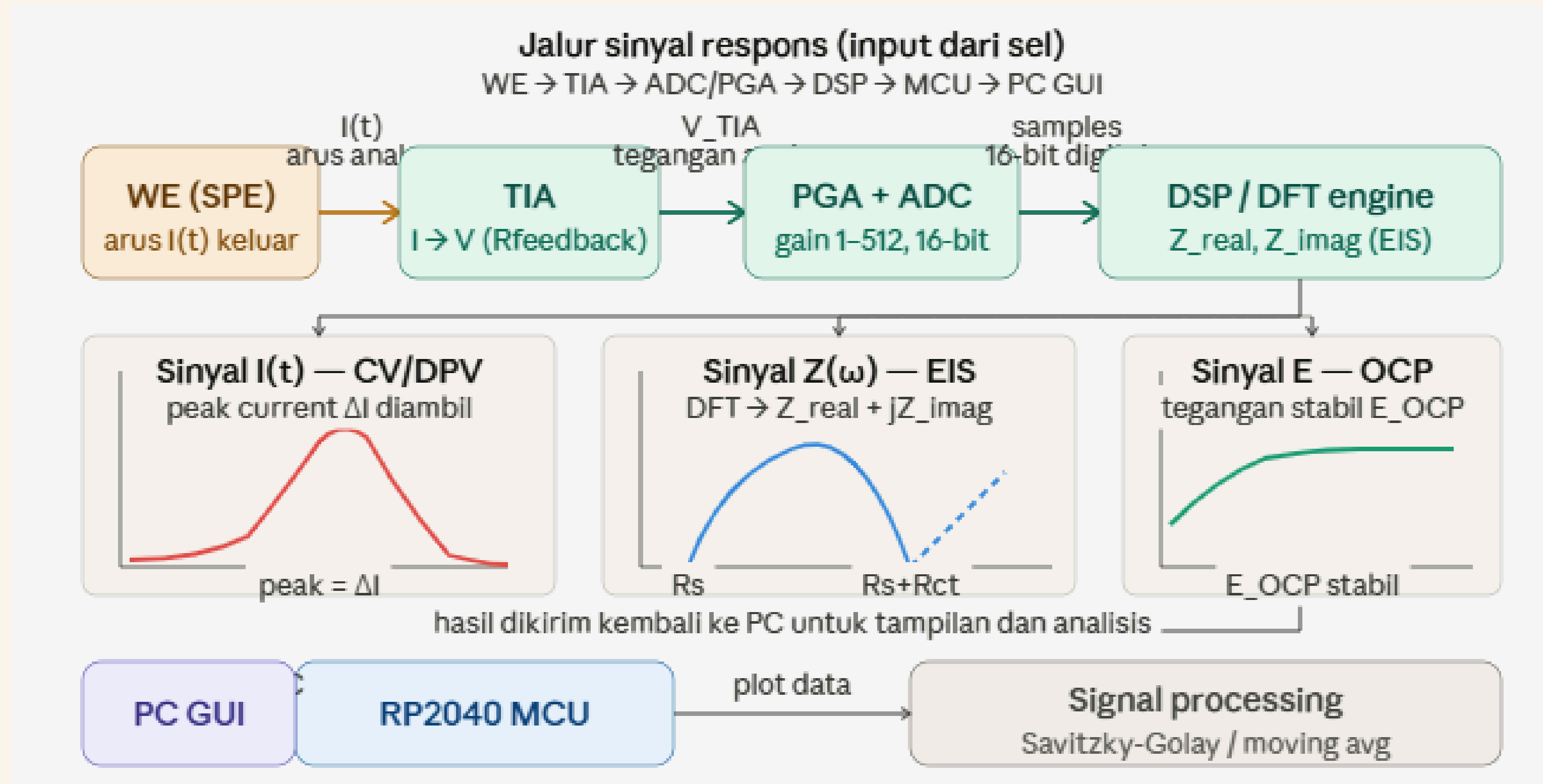


1. Dokumen (Diagram Blok)





1. Dokumen (Diagram Blok)



1. Dokument (RCAL)

RCAL	<code>tia_rf</code> (r)	<code>amplitude</code> (z)	<code>_CGmax</code> (t)	<code>_CGmin</code> (u)
3.3 Ω	<code>r0</code> (200 Ω)	<code>z20</code> ($\approx$ 8 mV)	1000	300
4.7 Ω	<code>r0</code> (200 Ω)	<code>z30</code> ($\approx$ 12 mV)	1500	400
200 Ω	<code>r1</code> (1 k Ω)	<code>z60</code> ($\approx$ 24 mV)	5000	1500
10 k Ω (default)	<code>r3</code> (10 k Ω)	<code>z126</code> ($\approx$ 50 mV)	30000	7500

1. Dokumen (RCAL)

RCAL	Rentang Z terukur (ideal)	Cocok untuk	Perintah serial
3.3 Ω	$\sim 0.3 \Omega - 330 \Omega$	Logam, elektroda karbon, baterai Li-ion ESR rendah	c3.3
4.7 Ω	$\sim 0.5 \Omega - 470 \Omega$	Baterai, superkapasitor, korosi aktif	c4.7
200 Ω	$\sim 20 \Omega - 20 \text{ k}\Omega$	Larutan elektrolit encer, biosensor, sensor pH	c200

1. Dokumen (RCAL)

200 Ω Rentang optimal: **20 Ω - 2 k Ω** Sweet spot: **100 - 1 k Ω**

c200

KONFIGURASI FIRMWARE YANG DISARANKAN

tia_rf (r)
r1 (1 k Ω)

amplitude (Y)
Y40-Y80 mV

_CGmax (t)
5000

_CGmin (u)
1500

APLIKASI YANG COCOK

- **Larutan elektrolit encer-sedang**

PBS 1 $\times$ (~100-300 Ω), NaCl 0.1 M (~200 Ω), medium kultur sel. Paling umum di lab bio.

- **Biosensor impedimetrik**

Deteksi antibodi/antigen via perubahan impedansi permukaan elektroda, tipikal 100-2000 Ω .

- **Sensor pH / ion-selektif**

Impedansi membran ISE dan glass pH electrode biasanya ratusan Ohm hingga beberapa k Ω .

- **Elektrodeposisi & plating**

Impedansi sel plating dalam larutan asam/basa encer cocok dengan range ini.

1. Dokumen (RCAL)

10 k Ω Rentang optimal: **1 k Ω – 100 k Ω** Sweet spot: **5 k Ω – 50 k Ω**

c10000 (default)

KONFIGURASI FIRMWARE YANG DISARANKAN

tia_rf (r)
r3 (10 k Ω)

amplitude (Y)
Y126 (50 mV)

_CGmax (t)
30000

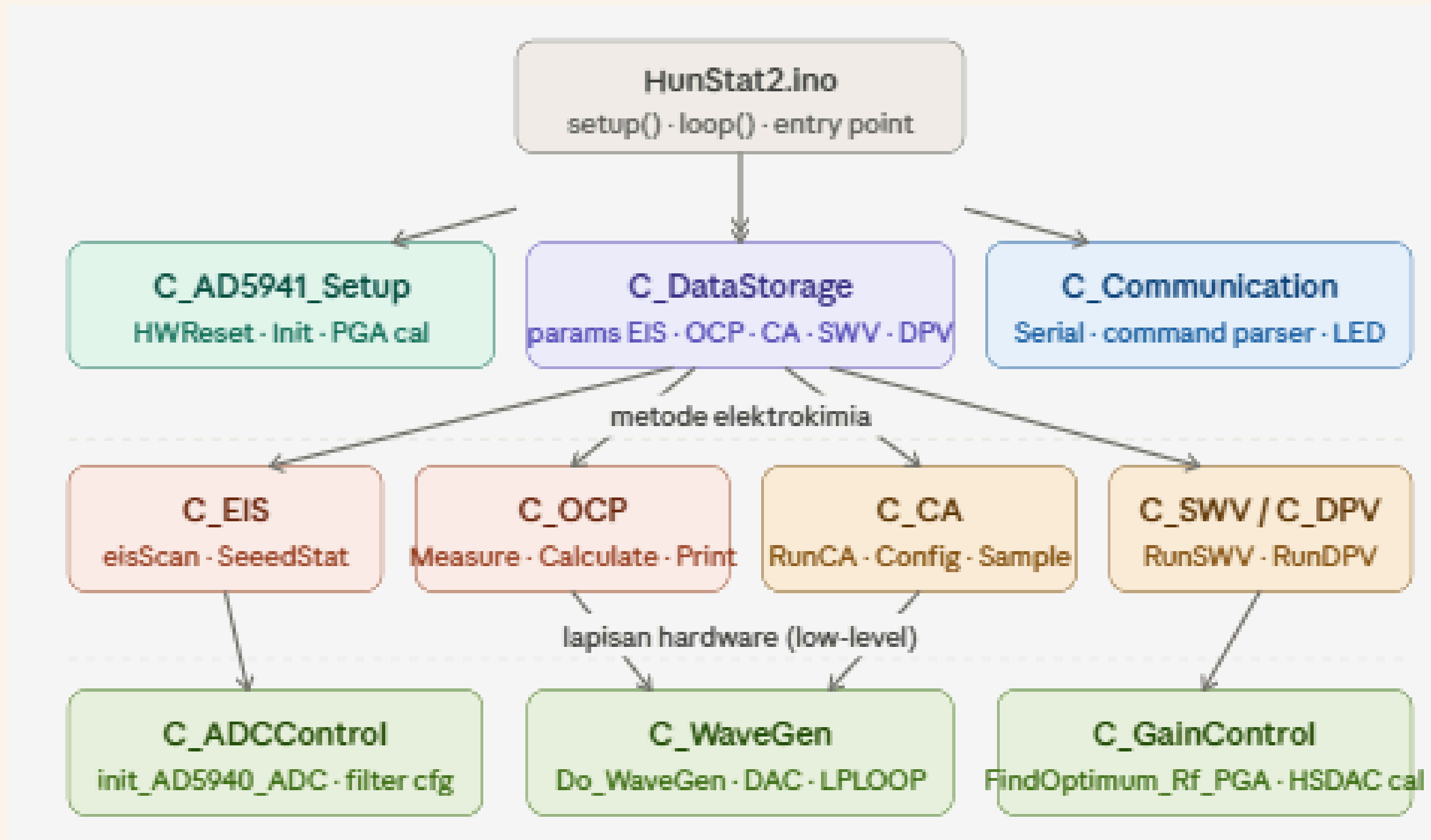
_CGmin (u)
7500

APLIKASI YANG COCOK

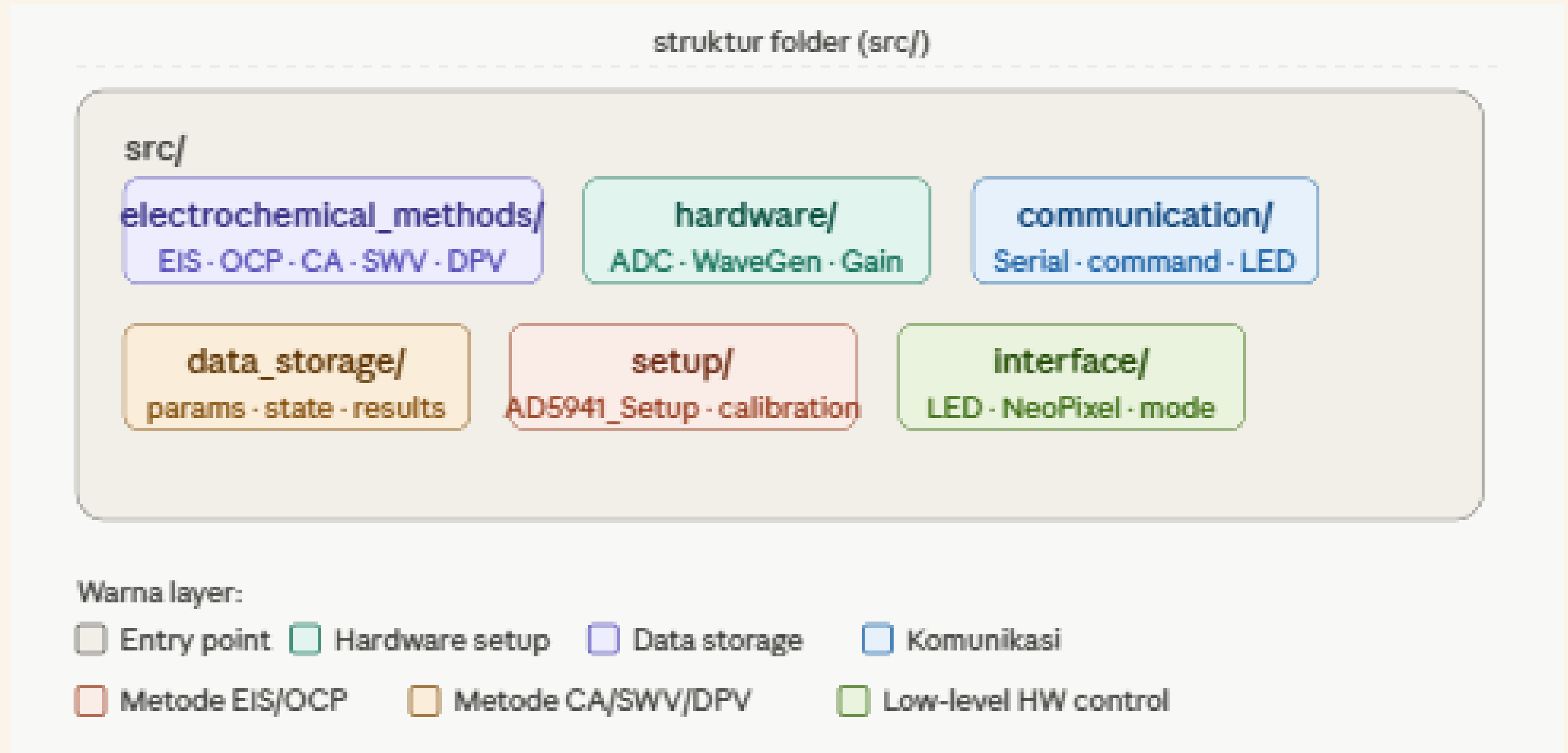
- **Larutan elektrolit sangat encer**
Aquades, air deionisasi, larutan < 1 mM. Impedansi R_s bisa > 10 k Ω .
- **Tanah / sedimen geokimia**
EIS untuk analisis tanah atau sedimen basah, impedansi tipikal 1–100 k Ω .

- **Polimer & membran**
Karakterisasi impedansi film polimer, membran Nafion, atau lapisan oksida tebal.
- **Semikonduktor & sel surya**
Karakterisasi interface junction semikonduktor, perovskite solar cell, DSSC.

2. Foldering Software



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```
HunStat2/
├── HunStat2.ino
├── src/
│   ├── setup/
│   │   ├── ad5941_setup.h/.cpp
│   │   └── ad5941_calibration.h
│   ├── hardware/
│   │   ├── adc_control.h/.cpp
│   │   ├── wave_gen.h/.cpp
│   │   └── gain_control.h/.cpp
│   ├── electrochemical_methods/
│   │   ├── c_eis.h/.cpp
│   │   ├── c_ocp.h/.cpp
│   │   ├── c_ca.h/.cpp
│   │   ├── c_swv.h/.cpp
│   │   └── c_dpv.h/.cpp
│   ├── data_storage/
│   │   └── data_storage.h/.cpp
│   ├── communication/
│   │   └── communication.h/.cpp
└── LED)
    └── interface/
        └── led_interface.h/.cpp
```

← entry point, semirip FreiStat.ino (bersih)

← C_AD5941_Setup (HWReset, Init, PGA cal)

← Calibrate_HSDAC, PGA cal

← C_ADCCControl (init_AD5940_ADC, filter cfg)

← C_WaveGen (Do_WaveGen, SetDACLevel, LPLOOP)

← C_GainControl (FindOptimum_Rf_PGA)

← C_EIS (eisScan, SeedStatScan, Nyquist)

← C_OCP (Config, Measure, Calculate, Print)

← C_CA (RunCA, Config_DCMeasurement)

← C_SWV (RunSWV)

← C_DPV (RunDPV)

← C_DataStorage (semua parameter global)

← C_Communication (Serial, command parser,

LED)

← C_LEDInterface (NeoPixel, LED warna)

3. Laporan Jasa Potensiostat

LAPORAN PENGAJUAN JASA

Program Riset

WCU Equity – Riset Internasional Unggul Top Tier 10% Tahun 2025



KODE : DRI.PN-17-131-2025

NO. KONTRAK : 60A/IT1.B07.1/KP/2026

JUDUL PENELITIAN : Pengembangan Sistem Potensiostat Berbasis AD5940 untuk Pengukuran Elektrokimia

JUDUL UNIT JASA : Asisten Riset

NO. INVOICE : –

TANGGAL : 30 Mei 2026

KETUA PENELITI : Isa Anshori, S.T., M.Eng., Ph.D.

AFILIASI : KK Teknik Biomedika, Sekolah Teknik Elektro dan Informatika, Institut Teknologi Bandung

